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Practice Paper 3: Mark Scheme and Worked Solutions

Pure Mathematics • Level: Challenge

How to use this mark scheme

Each question has a **mark scheme** in the style the exam boards use, then a **worked solution** that explains each step. Mark your own work with the scheme first, then read the worked solution for anything you missed.

M	Method mark: a correct method applied to the problem, even if there are small slips.
dM	Dependent method mark: needs the method mark before it.
A	Accuracy mark: a correct answer from correct working. Needs the method mark it follows.
B	Independent mark for a correct result or statement.
ft	Follow through: allows an earlier wrong answer to be used correctly.
cso	Correct solution only: every step must be correct, with no errors.
awrt	Answers which round to the value given.
oe	Or equivalent: any correct equivalent form is accepted.

For "show that" questions the answer is given, so every step must be shown for full marks. Accept any correct method that is not in the scheme, and give the marks it would earn.

Question 1

Part	Scheme	Marks	Notes
	Assume $\log_2 3 = \frac{p}{q}$, where p, q are positive integers	B1	Positive since $3 > 1$
	$2^{p/q} = 3 \Rightarrow 2^p = 3^q$	M1	Removes the log and the fraction
	2^p is even (as $p \geqslant 1$) but 3^q is odd	M1	Parity argument
	Contradiction, so $\log_2 3$ is irrational	A1	cso
		(4)	

Worked solution

Assume, for contradiction, that $\log_2 3$ is rational. Since $3 > 2^0 = 1$, $\log_2 3 > 0$, so we can write $\log_2 3 = \frac{p}{q}$ with p and q positive integers. Then

$$2^{p/q} = 3 \implies 2^p = 3^q$$

Since $p \geqslant 1$, 2^p is even. But 3^q is a product of odd numbers, so it is odd. An even number can't equal an odd number. This is a contradiction, so $\log_2 3$ is irrational.

Watch out: saying p and q are positive matters. If p could be 0, then $2^p = 1$ would be odd.

Question 2

Part	Scheme	Marks	Notes
	$\log_y x = \dfrac{1}{\log_x y}$, with a reason	M1	
	$t + \dfrac{1}{t} = \dfrac{5}{2} \Rightarrow 2t^2 - 5t + 2 = 0$	M1	Where $t = \log_x y$
	$t = 2$ or $t = \frac{1}{2}$	A1	
	$t = 2 \Rightarrow y = x^2 \Rightarrow x^3 = 27$	M1	Uses $xy = 27$
	$x = 3,\ y = 9$	A1	
	$x = 9,\ y = 3$	A1	
		(6)	

Worked solution

Let $t = \log_x y$, so $y = x^t$. Raising both sides to the power $\frac{1}{t}$ gives $x = y^{1/t}$, so $\log_y x = \frac{1}{t}$. The first equation becomes

$$t + \frac{1}{t} = \frac{5}{2} \implies 2t^2 - 5t + 2 = 0 \implies (2t - 1)(t - 2) = 0$$

- $t = 2$: $y = x^2$, so $xy = x^3 = 27$. Then $x = 3$ and $y = 9$.
- $t = \frac{1}{2}$: $y = x^{1/2}$, so $x = y^2$ and $xy = y^3 = 27$. Then $y = 3$ and $x = 9$.

The solutions are $x = 3, y = 9$ and $x = 9, y = 3$.

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Question 3

Part	Scheme	Marks	Notes
(a)	$(a+3d)^2 = a(a+12d)$	M1	Uses the 1st, 4th and 13th terms
	$9d^2 = 6ad \Rightarrow a = \frac{3}{2}d$ (as $d \neq 0$)	M1	
	$r = \dfrac{a+3d}{a} = 3$	A1	
(b)	$a(1+3+9+27) = 120 \Rightarrow a = 3$	M1	
	$d = 2$, $S_{20} = \frac{20}{2}(2\times 3 + 19\times 2)$	M1	
	440	A1	
		(6)	

Worked solution

(a) Let the arithmetic sequence have first term a and common difference d. Its 1st, 4th and 13th terms are a, $a+3d$ and $a+12d$. These are consecutive terms of a geometric sequence, so the middle one squared equals the product of the other two:

$$(a+3d)^2 = a(a+12d) \;\Rightarrow\; 6ad + 9d^2 = 12ad \;\Rightarrow\; 9d^2 = 6ad \;\Rightarrow\; a = \tfrac{3}{2}d$$

We can divide by d because $d \neq 0$. Then $r = \dfrac{a+3d}{a} = \dfrac{\frac{9}{2}d}{\frac{3}{2}d} = 3$.

(b) The geometric sequence starts at a with ratio 3, so $a + 3a + 9a + 27a = 40a = 120$ and $a = 3$. Then $d = \frac{2}{3}a = 2$, and

$$S_{20} = \tfrac{20}{2}\,(2(3) + 19(2)) = 10 \times 44 = 440$$

Check: the arithmetic sequence is $3, 5, 7, 9, \ldots$, with 4th term 9 and 13th term 27. So $3, 9, 27$ is geometric.

Question 4

Part	Scheme	Marks	Notes
(a)	$u_2 = \frac{1}{3},\ u_3 = -\frac{1}{2},\ u_4 = -3,\ u_5 = 2$	B1	
	Periodic with period 4	B1	
(b)	$25\left(2 + \frac{1}{3} - \frac{1}{2} - 3\right)$	M1	
	$-\frac{175}{6}$	A1	
(c)	Constant needs $u = \dfrac{u-1}{u+1}$	M1	
	$u^2 + u = u - 1 \Rightarrow u^2 = -1$	M1	
	No real solutions, so no constant sequence	A1	Needs a conclusion
		(7)	

Worked solution

(a) $u_2 = \frac{2-1}{2+1} = \frac{1}{3}$, $u_3 = \frac{-2/3}{4/3} = -\frac{1}{2}$, $u_4 = \frac{-3/2}{1/2} = -3$, $u_5 = \frac{-4}{-2} = 2$.
Since $u_5 = u_1$, every term after that repeats: the sequence is periodic with period 4.

(b) 100 terms is exactly 25 full cycles:

$$\sum_{r=1}^{100} u_r = 25\left(2 + \tfrac{1}{3} - \tfrac{1}{2} - 3\right) = 25 \times \left(-\tfrac{7}{6}\right) = -\tfrac{175}{6}$$

(c) If the sequence is constant, then $u_2 = u_1 = u$, so

$$u = \frac{u-1}{u+1} \;\Rightarrow\; u^2 + u = u - 1 \;\Rightarrow\; u^2 = -1$$

No real number squares to -1, so no real u_1 gives a constant sequence. ($u_1 = -1$ isn't possible either, since u_2 would be undefined.)

Question 5

Part	Scheme	Marks	Notes
	$I = \mathrm{e}^{2x}\sin x - \int 2\mathrm{e}^{2x}\sin x\,\mathrm{d}x$	M1	Parts once
	$\int \mathrm{e}^{2x}\sin x\,\mathrm{d}x = -\mathrm{e}^{2x}\cos x + \int 2\mathrm{e}^{2x}\cos x\,\mathrm{d}x$	M1	Parts again, same choice of u
	$I = \mathrm{e}^{2x}\sin x + 2\mathrm{e}^{2x}\cos x - 4I$	M1	Recognises I
	$I = \frac{1}{5}\mathrm{e}^{2x}(\sin x + 2\cos x)\ (+c)$	A1	
	$\frac{1}{5}\mathrm{e}^{\pi}(1+0) - \frac{1}{5}(0+2)$	dM1	Correct limits
	$\dfrac{\mathrm{e}^{\pi}-2}{5}$	A1	
		(6)	

Worked solution

Let $I = \int \mathrm{e}^{2x}\cos x\,\mathrm{d}x$. Use parts with $u = \mathrm{e}^{2x}$ both times, so the integral comes back to I.
First, with $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$:

$$I = \mathrm{e}^{2x}\sin x - \int 2\mathrm{e}^{2x}\sin x\,\mathrm{d}x$$

Second, with $\frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$:

$$\int \mathrm{e}^{2x}\sin x\,\mathrm{d}x = -\mathrm{e}^{2x}\cos x + \int 2\mathrm{e}^{2x}\cos x\,\mathrm{d}x = -\mathrm{e}^{2x}\cos x + 2I$$

Substitute back:

$$I = \mathrm{e}^{2x}\sin x - 2\left(-\mathrm{e}^{2x}\cos x + 2I\right) \;\Rightarrow\; 5I = \mathrm{e}^{2x}(\sin x + 2\cos x)$$

$$\int_0^{\pi/2} \mathrm{e}^{2x}\cos x\,\mathrm{d}x = \left[\tfrac{1}{5}\mathrm{e}^{2x}(\sin x + 2\cos x)\right]_0^{\pi/2} = \frac{\mathrm{e}^{\pi}}{5} - \frac{2}{5} = \frac{\mathrm{e}^{\pi}-2}{5}$$

Watch out: if you swap the choice of u the second time, you undo the first step and end up with $I = I$.

Question 6

Part	Scheme	Marks	Notes
	$\sin 2x = \sin x \Rightarrow \sin x(2\cos x - 1) = 0$	M1	
	$x = 0, \frac{\pi}{3}, \pi$	A1	
	$\int_0^{\pi/3} (\sin 2x - \sin x)\,\mathrm{d}x + \int_{\pi/3}^{\pi} (\sin x - \sin 2x)\,\mathrm{d}x$	M1	Splits at $\frac{\pi}{3}$, right way round
	$\pm\left(-\frac{1}{2}\cos 2x + \cos x\right)$	M1	Integrates both terms
	First region $\frac{1}{4}$	A1	
	Second region $\frac{9}{4}$	A1	
	Total area $\frac{5}{2}$	A1	
		(7)	

Worked solution

The curves meet where $2\sin x\cos x = \sin x$, so $\sin x(2\cos x - 1) = 0$. In $[0, \pi]$ this gives $x = 0, \frac{\pi}{3}, \pi$. At $x = \frac{\pi}{6}$, $\sin 2x = 0.87 > \sin x = 0.5$, so $\sin 2x$ is on top in $\left(0, \frac{\pi}{3}\right)$. At $x = \frac{\pi}{2}$, $\sin x = 1 > \sin 2x = 0$, so $\sin x$ is on top in $\left(\frac{\pi}{3}, \pi\right)$.

$$\int_0^{\pi/3} (\sin 2x - \sin x)\,\mathrm{d}x = \left[-\tfrac{1}{2}\cos 2x + \cos x\right]_0^{\pi/3} = \left(\tfrac{1}{4} + \tfrac{1}{2}\right) - \left(-\tfrac{1}{2} + 1\right) = \tfrac{1}{4}$$

$$\int_{\pi/3}^{\pi} (\sin x - \sin 2x)\,\mathrm{d}x = \left[-\cos x + \tfrac{1}{2}\cos 2x\right]_{\pi/3}^{\pi} = \left(1 + \tfrac{1}{2}\right) - \left(-\tfrac{1}{2} - \tfrac{1}{4}\right) = \tfrac{9}{4}$$

Total area $= \frac{1}{4} + \frac{9}{4} = \frac{5}{2}$.

Watch out: integrating $\sin 2x - \sin x$ from 0 to π in one go gives -2. The regions above and below cancel, so it is not the area.

Question 7

Part	Scheme	Marks	Notes
(a)	Centre (r, r), radius r (first quadrant, as P is)	B1	
	$(9 - r)^2 + (2 - r)^2 = r^2$	M1	
	$r^2 - 22r + 85 = 0$	A1	
	$r = 5$ or $r = 17$	M1	
	$(x - 5)^2 + (y - 5)^2 = 25$ and $(x - 17)^2 + (y - 17)^2 = 289$	A1	Both
(b)	Both circles are symmetric in $y = x$, or solve them simultaneously	M1	
	$Q = (2, 9)$	A1	
	$PQ = 7\sqrt{2}$	A1	
		(8)	

Worked solution

(a) A circle touching both axes has its centre the same distance r from each axis, and that distance is

the radius. P is in the first quadrant, so the centre is (r, r). P is on the circle:

$$(9-r)^2+(2-r)^2=r^2 \;\Rightarrow\; r^2-22r+85=0 \;\Rightarrow\; (r-5)(r-17)=0$$

The circles are $(x-5)^2+(y-5)^2=25$ and $(x-17)^2+(y-17)^2=289$.
(b) Both centres lie on $y=x$, so both circles are symmetric in that line. Reflecting $P(9,2)$ in $y=x$ gives $(2,9)$, which therefore lies on both circles: $Q=(2,9)$.
You can also expand both equations and subtract them, which gives $x+y=11$. Substituting back gives $x^2-11x+18=0$, so $x=2$ or 9.

$$PQ=\sqrt{(9-2)^2+(2-9)^2}=\sqrt{98}=7\sqrt{2}$$

Question 8

Part	Scheme	Marks	Notes
	$\frac{dy}{dx}=-\frac{4}{x^2}=-\frac{1}{p^2}$ at P	M1	
	$y-\frac{2}{p}=-\frac{1}{p^2}(x-2p)$	M1	Tangent through P
	$A(4p,0)$ and $B\left(0,\frac{4}{p}\right)$	A1	
	Area $=\frac{1}{2}\times 4p\times\frac{4}{p}$	M1	
	$=8$, which doesn't depend on p	A1	
	Midpoint of AB is $\left(2p,\frac{2}{p}\right)=P$	B1	
		(6)	

Worked solution

$y=\frac{4}{x}$, so $\frac{dy}{dx}=-\frac{4}{x^2}$. At $x=2p$ the gradient is $-\frac{4}{4p^2}=-\frac{1}{p^2}$. The tangent at P is

$$y-\frac{2}{p}=-\frac{1}{p^2}(x-2p)$$

At A, $y=0$: $-\frac{2}{p}=-\frac{x-2p}{p^2}$, so $x-2p=2p$ and $x=4p$.
At B, $x=0$: $y=\frac{2}{p}+\frac{2p}{p^2}=\frac{4}{p}$.
The triangle OAB has a right angle at O, so

$$\text{Area}=\frac{1}{2}\times 4p\times\frac{4}{p}=8$$

for every $p>0$. The midpoint of $A(4p,0)$ and $B\left(0,\frac{4}{p}\right)$ is $\left(2p,\frac{2}{p}\right)$, which is P.

Question 9

Part	Scheme	Marks	Notes
	Vertices A, B, C with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$; midpoint of BC is $\frac{1}{2}(\mathbf{b}+\mathbf{c})$	B1	
	Point on the median from A: $\mathbf{a} + \lambda\left(\frac{1}{2}(\mathbf{b}+\mathbf{c}) - \mathbf{a}\right)$	M1	
	$\lambda = \frac{2}{3}$ gives $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$	A1	
	By symmetry, or by checking, the same point lies on the other two medians	M1	
	So all three medians pass through $G = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$	A1	
	$\lambda = \frac{2}{3}$, so G divides each median in the ratio $2:1$	A1	Needs a conclusion
		(6)	

Worked solution

Let the triangle have vertices A, B, C with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Let L, M, N be the midpoints of BC, CA, AB. Then $\overrightarrow{OL} = \frac{1}{2}(\mathbf{b}+\mathbf{c})$, and similarly for M and N.
The point $\frac{2}{3}$ of the way along AL from A is

$$\mathbf{a} + \frac{2}{3}\left(\frac{1}{2}(\mathbf{b}+\mathbf{c}) - \mathbf{a}\right) = \frac{1}{3}\mathbf{a} + \frac{1}{3}\mathbf{b} + \frac{1}{3}\mathbf{c}$$

This expression treats $\mathbf{a}, \mathbf{b}, \mathbf{c}$ the same way. Swapping the letters gives the point $\frac{2}{3}$ of the way along BM from B, and along CN from C, and the expression doesn't change. So all three medians pass through

$$G = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$$

On each median, G is $\frac{2}{3}$ of the way from the vertex, so it divides each median in the ratio $2:1$.

Another way: write a general point on two medians with parameters λ and μ, equate coefficients of two non-parallel vectors such as $\mathbf{b}-\mathbf{a}$ and $\mathbf{c}-\mathbf{a}$, and find $\lambda = \mu = \frac{2}{3}$. Then show G lies on the third median.

Question 10

Part	Scheme	Marks	Notes
(a)	$\cos(2\theta+\theta) = \cos 2\theta\cos\theta - \sin 2\theta\sin\theta$	M1	
	$= (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta$, with $\sin^2\theta = 1 - \cos^2\theta$	M1	
	$= 4\cos^3\theta - 3\cos\theta$	A1*	cso
(b)	$x = \cos\theta$: $2(4\cos^3\theta - 3\cos\theta) = 1 \Rightarrow \cos 3\theta = \frac{1}{2}$	M1	
	$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$	M1	Three values
	$x = \cos\frac{\pi}{9},\ \cos\frac{5\pi}{9},\ \cos\frac{7\pi}{9}$	A1	
	These are distinct, and a cubic has at most three roots	A1	
(c)	$8(x-x_1)(x-x_2)(x-x_3)$: constant term $-8x_1x_2x_3 = -1$	M1	
	$\frac{1}{8}$	A1	
		(9)	

Worked solution

(a)

$$\begin{aligned}\cos 3\theta = \cos(2\theta+\theta) &= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta\\ &= (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta\\ &= 2\cos^3\theta - \cos\theta - 2(1-\cos^2\theta)\cos\theta = 4\cos^3\theta - 3\cos\theta\end{aligned}$$

(b) Put $x = \cos\theta$. Then $8x^3 - 6x - 1 = 2(4\cos^3\theta - 3\cos\theta) - 1 = 2\cos 3\theta - 1$, so the equation becomes $\cos 3\theta = \frac{1}{2}$.
Take $0 \leqslant \theta \leqslant \pi$, so $0 \leqslant 3\theta \leqslant 3\pi$. Then $3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$, and $\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$.
Cos is decreasing on $[0,\pi]$, so $\cos\frac{\pi}{9}$, $\cos\frac{5\pi}{9}$ and $\cos\frac{7\pi}{9}$ are three different numbers. A cubic has at most three roots, so these are all of them:

$$x = \cos\frac{\pi}{9}, \quad \cos\frac{5\pi}{9}, \quad \cos\frac{7\pi}{9}$$

(c) Call the roots x_1, x_2, x_3. Then $8x^3 - 6x - 1 \equiv 8(x-x_1)(x-x_2)(x-x_3)$. Comparing constant terms gives $-1 = -8x_1x_2x_3$, so

$$\cos\frac{\pi}{9}\cos\frac{5\pi}{9}\cos\frac{7\pi}{9} = \frac{1}{8}$$

Question 11

Part	Scheme	Marks	Notes
(a)	$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$	M1	Implicit differentiation, product rule on xy
	$\frac{dy}{dx} = -\frac{2x+y}{x+2y}$	A1	
	$\frac{dy}{dx} = 0 \Rightarrow y = -2x$	M1	
	$x^2 - 2x^2 + 4x^2 = 12 \Rightarrow x = \pm 2$	M1	
	$(2,-4)$ and $(-2,4)$	A1	
(b)	A valid method: normal through O, or bounding x^2+y^2 algebraically	M1	
	$y^2 = x^2$, or $(x+y)^2 = 12 + xy \geqslant 0$	M1	
	Points $\left(2\sqrt{3}, -2\sqrt{3}\right)$ and $\left(-2\sqrt{3}, 2\sqrt{3}\right)$ at distance $\sqrt{24}$	A1	
	Shows this is the greatest, for example the other points are at distance $\sqrt{8}$	M1	
	$2\sqrt{6}$	A1	
		(10)	

Worked solution

(a) Differentiate implicitly, using the product rule on xy:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \;\Rightarrow\; \frac{dy}{dx} = -\frac{2x+y}{x+2y}$$

This is zero when $y = -2x$. Substitute: $x^2 - 2x^2 + 4x^2 = 3x^2 = 12$, so $x = \pm 2$. The points are $(2,-4)$ and $(-2,4)$, and the denominator $x + 2y = \mp 6 \neq 0$ at both.

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(b) *Method 1 (algebra).* From the curve, $x^2 + y^2 = 12 - xy$. Also

$$(x+y)^2 = x^2 + 2xy + y^2 = 12 + xy \geqslant 0 \implies xy \geqslant -12 \implies x^2 + y^2 = 12 - xy \leqslant 24$$

Equality needs $x + y = 0$. Then $x^2 = 12$, giving the points $\left(2\sqrt{3}, -2\sqrt{3}\right)$ and $\left(-2\sqrt{3}, 2\sqrt{3}\right)$. The greatest distance is $\sqrt{24} = 2\sqrt{6}$.

Method 2 (calculus). At the farthest point, the tangent is perpendicular to the line from O. So $\frac{y}{x} \times \left(-\frac{2x+y}{x+2y}\right) = -1$, which simplifies to $y^2 = x^2$.

- $y = x$: $3x^2 = 12$, giving $(\pm 2, \pm 2)$ at distance $\sqrt{8}$ (the closest points).
- $y = -x$: $x^2 = 12$, giving distance $\sqrt{24}$.

The greatest distance is $2\sqrt{6}$.

Question 12

Part	Scheme	Marks	Notes
(a)	$\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \ln x}{x^2}$	M1	Quotient or product rule
	$= \frac{1 - \ln x}{x^2}$	A1	
	$= 0$ only when $\ln x = 1$: one stationary point, $\left(e, \frac{1}{e}\right)$	A1	
	Sign of $\frac{dy}{dx}$ changes from + to −, so it is a maximum	A1	Or second derivative
(b)	$\pi > e$, so $\frac{\ln \pi}{\pi} < \frac{1}{e}$	M1	Uses the unique maximum
	$e \ln \pi < \pi \Rightarrow \ln(\pi^e) < \ln(e^\pi)$	M1	
	ln is increasing, so $\pi^e < e^\pi$	A1	cso
		(7)	

Worked solution

(a) By the quotient rule,

$$\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

This is zero only when $\ln x = 1$, so $x = e$. There is exactly one stationary point, $\left(e, \frac{1}{e}\right)$.

For $x < e$, $\ln x < 1$, so $\frac{dy}{dx} > 0$. For $x > e$, $\frac{dy}{dx} < 0$. So it is a **maximum**, and the curve is decreasing for all $x > e$.

(b) $\pi \neq e$ and $\left(e, \frac{1}{e}\right)$ is the only maximum, so

$$\frac{\ln \pi}{\pi} < \frac{1}{e} \implies e \ln \pi < \pi \implies \ln(\pi^e) < \ln(e^\pi)$$

ln is an increasing function, so $\pi^e < e^\pi$. (On a calculator, $e^\pi = 23.14\ldots$ and $\pi^e = 22.46\ldots$)

Watch out: a calculator check is not a proof. The marks are for the argument.

Question 13

Part	Scheme	Marks	Notes
	$(1-x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)(-x) + \dfrac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-x)^2 + \ldots$	M1	
	$= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \ldots$	A1	
	Compares with $S = 1 + \frac{1}{4} + \frac{3}{32} + \frac{5}{128} + \ldots$	M1	
	Spots $x = \frac{1}{2}$: $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, $\frac{3}{8} \cdot \frac{1}{4} = \frac{3}{32}, \ldots$	M1	
	General term: coefficient of x^r is $\dfrac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!}$	M1	
	Term of S is $\dfrac{1 \times 3 \times \cdots \times (2r-1)}{4^r\, r!}$, which is that coefficient times $\left(\frac{1}{2}\right)^r$	A1	
	Valid, as $\left\lvert\frac{1}{2}\right\rvert < 1$	B1	
	$S = \left(1 - \frac{1}{2}\right)^{-\frac{1}{2}} = \sqrt{2}$	A1	
		(8)	

Worked solution

The expansion of $(1-x)^{-\frac{1}{2}}$ is

$$1 + \frac{1}{2}x + \frac{\frac{1}{2}\cdot\frac{3}{2}}{2!}x^2 + \frac{\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2}}{3!}x^3 + \cdots = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \ldots$$

The minus signs in $-\frac{1}{2}$ and in $-x$ cancel in pairs, so every term is positive. In general, the coefficient of x^r is

$$\frac{\frac{1}{2}\cdot\frac{3}{2}\cdots\frac{2r-1}{2}}{r!} = \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!}$$

The denominators in S are $4 \times 8 \times \cdots \times 4r = 4^r\, r!$, so the rth term of S is

$$\frac{1 \times 3 \times \cdots \times (2r-1)}{4^r\, r!} = \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!} \times \left(\frac{1}{2}\right)^r$$

This is the binomial term with $x = \frac{1}{2}$. The expansion is valid because $\left\lvert\frac{1}{2}\right\rvert < 1$, so

$$S = \left(1 - \frac{1}{2}\right)^{-\frac{1}{2}} = \left(\frac{1}{2}\right)^{-\frac{1}{2}} = \sqrt{2}$$

Check: the first five terms add up to 1.3999, which is approaching $\sqrt{2} = 1.4142$.

Question 14

Part	Scheme	Marks	Notes
(a)	$\displaystyle\int \frac{10}{1-\sqrt{h}}\,\mathrm{d}h = \int 1\,\mathrm{d}t$	M1	Separates the variables
	$h = u^2,\ \mathrm{d}h = 2u\,\mathrm{d}u$	B1	
	$\displaystyle\int \frac{20u}{1-u}\,\mathrm{d}u$	M1	Fully in terms of u
	$\displaystyle\frac{20u}{1-u} = -20 + \frac{20}{1-u}$	M1	Divides, or uses another valid method
	$-20u - 20\ln(1-u)$	A1	
	$t = 0,\ h = 0 \Rightarrow c = 0$	M1	
	$t = -20\left(\sqrt{h} + \ln\left(1-\sqrt{h}\right)\right)$	A1*	cso
(b)	$t = -20(0.9 + \ln 0.1)$	M1	
	28.1 minutes	A1	awrt 28.1
(c)	As $h \to 1$, $\ln(1-\sqrt{h}) \to -\infty$, so $t \to \infty$	B1	oe, $\frac{\mathrm{d}h}{\mathrm{d}t} \to 0$
		(10)	

Worked solution

(a) Separate the variables:

$$\int \frac{10}{1-\sqrt{h}}\,\mathrm{d}h = \int 1\,\mathrm{d}t$$

Let $u = \sqrt{h}$, so $h = u^2$ and $\mathrm{d}h = 2u\,\mathrm{d}u$:

$$\int \frac{10}{1-u}\cdot 2u\,\mathrm{d}u = \int \frac{20u}{1-u}\,\mathrm{d}u$$

Write the numerator in terms of $1-u$: $20u = 20 - 20(1-u)$, so $\dfrac{20u}{1-u} = \dfrac{20}{1-u} - 20$. Then

$$\int \left(\frac{20}{1-u} - 20\right)\mathrm{d}u = -20\ln(1-u) - 20u + c$$

Here $0 \leqslant u < 1$, so $1-u > 0$ and no modulus sign is needed. So $t = -20\sqrt{h} - 20\ln\left(1-\sqrt{h}\right) + c$. At $t = 0$, $h = 0$, which gives $0 = 0 - 20\ln 1 + c$, so $c = 0$:

$$t = -20\left(\sqrt{h} + \ln\left(1-\sqrt{h}\right)\right)$$

(b) $h = 0.81$ gives $\sqrt{h} = 0.9$, so $t = -20(0.9 + \ln 0.1) = -20(0.9 - 2.3026) = 28.1$ minutes.

(c) As $h \to 1$, $1-\sqrt{h} \to 0$, so $\ln\left(1-\sqrt{h}\right) \to -\infty$ and $t \to \infty$. The depth gets closer and closer to 1 m but never reaches it. Equivalently, the rate $\frac{\mathrm{d}h}{\mathrm{d}t}$ tends to 0 as h approaches 1.

Watch out: $\frac{\mathrm{d}}{\mathrm{d}u}\ln(1-u) = -\frac{1}{1-u}$, so $\int \frac{20}{1-u}\,\mathrm{d}u = -20\ln(1-u)$. The minus sign is easy to lose.

End of mark scheme. Every answer here was checked with computer algebra.

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